

INTRODUCTION TO DATA SCIENCE

Data scientist is someone who extracts insights from messy data.

Facebook asks you to list your hometown and your current location, ostensibly to make it easier for your friends to find and connect with you.

But it also analyzes these locations to identify global migration patterns and where the fanbases of different football teams live.

As a large retailer, Target tracks your purchases and interactions, both online and in-store. And it uses the data to predictively model which of its customers are pregnant, to better market baby-related purchases to them.

In 2012, the Obama campaign employed dozens of data scientists who data mined and experimented their way to identifying voters who needed extra attention, choosing optimal donor-specific fundraising appeals and programs, and focusing get-out-the-vote efforts where they were most likely to be useful. And in 2016 the Trump campaign tested a staggering variety of online ads and analyzed the data to find what worked and what didn't.

some data scientists also occasionally use their skills for good—using data to make government more effective, to help the homeless, and to improve public health.

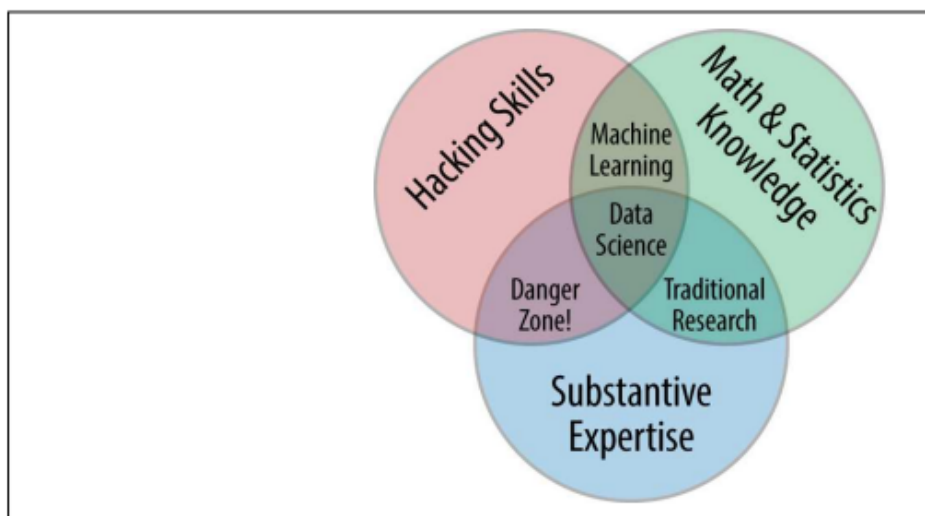


Figure P-1. Drew Conway's Data Science Venn Diagram

Data science comprises three distinct and overlapping areas: the skills of a statistician who knows how to model and summarize datasets (which are growing ever larger);

the skills of a computer scientist who can design and use algorithms to efficiently store, process, and visualize this data;

and the domain expertise—what we might think of as “classical” training in a subject—necessary both to formulate the right questions and to put their answers in context.

Big data is a blanket term for any collection of data sets so large or complex that it becomes difficult to process them using traditional data management techniques such as, for example, the RDBMS (relational database management systems).

The widely adopted RDBMS has long been regarded as a one-size-fits-all solution, but the demands of handling big data have shown otherwise.

Data science involves using methods to analyze massive amounts of data and extract the knowledge it contains.

You can think of the relationship between big data and data science as being like the relationship between crude oil and an oil refinery.

Data science and big data evolved from statistics and traditional data management but are now considered to be distinct disciplines.

The characteristics of big data are often referred to as the three Vs:

- *Volume—How much data is there?*
- *Variety—How diverse are different types of data?*
- *Velocity—At what speed is new data generated?*

Often these characteristics are complemented with a fourth V, veracity: How accurate is the data?

These four properties make big data different from the data found in traditional data management tools. Consequently, the challenges they bring can be felt in almost every aspect: data capture, curation, storage, search, sharing, transfer, and visualization.

In addition, big data calls for specialized techniques to extract the insights.

Data science is an evolutionary extension of statistics capable of dealing with the massive amounts of data produced today.

It adds methods from computer science to the repertoire of statistics.

*In a research note from Laney and Kart, *Emerging Role of the Data Scientist and the Art of Data Science*, the authors sifted through hundreds of job descriptions for data scientist, statistician, and BI (Business Intelligence) analyst to detect the differences between those titles. The main things that set a data scientist apart from a statistician are the ability to work with big data and experience in machine learning, computing, and algorithm building. Their tools tend to differ too, with data scientist job descriptions more frequently mentioning the ability to use Hadoop, Pig, Spark, R, Python, and Java, among others.*

Python is a great language for data science because it has many data science libraries available, and it's widely supported by specialized software.

For instance, almost every popular NoSQL database has a Python-specific API. Because of these features and the ability to prototype quickly with Python while keeping acceptable performance, its influence is steadily growing in the data science world.

Benefits and uses of data science and big data

Data science and big data are used almost everywhere in both commercial and non commercial settings. The number of use cases is vast, and the examples we'll provide throughout this book only scratch the surface of the possibilities.

Commercial companies in almost every industry use data science and big data to gain insights into their customers, processes, staff, completion, and products. Many companies use data science to offer customers a better user experience, as well as to cross-sell, up-sell, and personalize their offerings.

A good example of this is Google AdSense, which collects data from internet users so relevant commercial messages can be matched to the person browsing the internet.

Financial institutions use data science to predict stock markets, determine the risk of lending money, and learn how to attract new clients for their services.

Governmental organizations are also aware of data's value. Many governmental organizations not only rely on internal data scientists to discover valuable information, but also share their data with the public. You can use this data to gain insights or build data-driven applications.

Data.gov is but one example; it's the home of the US Government's open data. A data scientist in a governmental organization gets to work on diverse projects such as detecting fraud and other criminal activity or optimizing project funding.

A well-known example was provided by Edward Snowden, who leaked internal documents of the American National Security Agency and the British Government Communications Headquarters that show clearly how they used data science and big data to monitor millions of individuals.

Those organizations collected 5 billion data records from widespread applications such as Google Maps, Angry Birds, email, and text messages, among many other data sources. Then they applied data science techniques to distill information.

Nongovernmental organizations (NGOs) are also no strangers to using data. They use it to raise money and defend their causes. The World Wildlife Fund (WWF), for instance, employs data scientists to increase the effectiveness of their fundraising efforts.

Many data scientists devote part of their time to helping NGOs, because NGOs often lack the resources to collect data and employ data scientists.

DataKind is one such data scientist group that devotes its time to the benefit of mankind.

Universities use data science in their research but also to enhance the study experience of their students. The rise of massive open online courses (MOOC) produces a lot of data, which allows universities to study how this type of learning can complement traditional classes.

MOOCs are an invaluable asset if you want to become a data scientist and big data professional, so definitely look at a few of the better-known ones:

Coursera, Udacity, and edX. The big data and data science landscape changes quickly, and MOOCs allow you to stay up to date by following courses from top universities.

Data Science is a combination of multiple disciplines that uses statistics, data analysis, and machine learning to analyze data and to extract knowledge and insights from it.

Data science combines math and statistics, specialized programming, advanced analytics, artificial intelligence (AI) and machine learning with specific subject matter expertise to uncover actionable insights hidden in an organization's data. These insights can be used to guide decision making and strategic planning.

Data Science Tools

- Data Analysis: SAS, Jupyter, R Studio, MATLAB, Excel, RapidMiner
- Data Warehousing: Informatica/ Talend, AWS Redshift
- Data Visualization: Jupyter, Tableau, Cognos, RAW
- Machine Learning: Spark MLlib, Mahout, Azure ML studio

Applications of Data Science

- Healthcare
- Gaming
- Image Recognition
- Recommendation Systems · Logistics

- Fraud Detection
- Internet Search
- Speech Recognition
- Targeted Advertising
- Airline Route Planning
- Augmented Reality

FACETS OF DATA

1. Structured
2. Unstructured
3. Natural Language
4. Machine-generated
5. Graph-based
6. Audio, video, and images
7. Streaming

1. Structured

- o Depends on a data model
- o Resides in a fixed field within a record
- o SQL, or Structured Query Language, is the preferred way to manage and query data that resides in databases
- o Hierarchical Data

Indicator ID	Dimension List	Timeframe	Numeric Value	Missing Value Flag	Confidence Int
2	214390830 Total (Age-adjusted)	2008	74.6%		73.8%
3	214390833 Aged 18-44 years	2008	59.4%		58.0%
4	214390831 Aged 18-24 years	2008	17.4%		16.6%
5	214390822 Aged 25-44 years	2008	66.9%		65.3%
6	214390836 Aged 45-64 years	2008	58.6%		57.7%
7	214390834 Aged 45-54 years	2008	66.2%		65.1%
8	214390835 Aged 55-64 years	2008	91.5%		90.6%
9	214390840 Aged 65 years and over	2008	84.6%		83.8%
10	214390837 Aged 65-74 years	2008	93.6%		92.4%
11	214390838 Aged 75-84 years	2008	95.6%		94.4%
12	214390839 Aged 85 years and over	2008	98.0%		94.0%
13	214390841 Male (Age-adjusted)	2008	72.2%		71.1%
14	214390842 Female (Age-adjusted)	2008	76.8%		75.5%
15	214390843 White only (Age-adjusted)	2008	73.8%		72.9%
16	214390844 Black or African American only (Age-adjusted)	2008	77.6%		75.0%
17	214390845 American Indian or Alaska Native only (Age-adjusted)	2008	66.5%		57.1%
18	214390846 Asian only (Age-adjusted)	2008	80.5%		77.7%
19	214390847 Native Hawaiian or Other Pacific Islander only (Age-adjusted)	2008	0%		0%
20	214390848 2 or more races (Age-adjusted)	2008	73.6%		68.6%

Figure: An Excel table is an example of structured data.

2. Unstructured

- o Isn't easy to fit into a data model
- o Content is context-specific or

varying



o Example: email

Figure: Email is simultaneously an example of unstructured data and natural language data

2. Natural Language

- o It requires knowledge of specific data science techniques and linguistics
- o Entity Recognition, Topic Recognition, Summarization, Text Completion, and Sentiment Analysis
- o Example: email

2. Machine-generated

- o Automatically created by a computer, process, application, or other machine without human intervention data

3. Graph-based

- o Mathematical structure to model pair-wise relationships between objects.
- o Graph or network data that focuses on the relationship or adjacency of objects.
- o Graph structures use nodes, edges, and properties to represent and store graphical data.
- o Graph-based data is a natural way to represent social networks, and its structure allows you to calculate specific metrics such as the influence of a person and the shortest path between two people.
- o Graph databases are used to store graph-based data and are queried with specialized query languages such as SPARQL

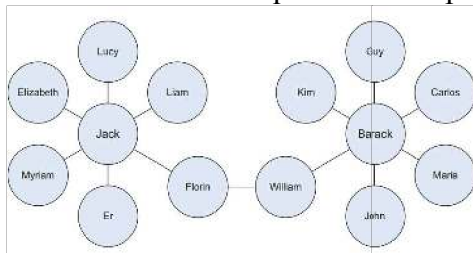


Figure: Friends in a social network are an example of graph-based data

6. Audio, Video, and Images

Audio, image, and video are data types that pose specific challenges to a data scientist.

```

CSIPERF:TXCOMMIT;313236
2014-11-28 11:36:13, Info
69), objectname [6]"(null)"
2014-11-28 11:36:13, Info
result 0x00000000, handle @0x4e54
2014-11-28 11:36:13, Info
Beginning NT transaction commit...
2014-11-28 11:36:13, Info
trace:
CSIPERF:TXCOMMIT;273983
2014-11-28 11:36:13, Info
70), objectname [6]"(null)"
2014-11-28 11:36:13, Info
result 0x00000000, handle @0x4e5c
2014-11-28 11:36:13, Info
Beginning NT transaction commit...
2014-11-28 11:36:14, Info
trace:
CSIPERF:TXCOMMIT;386259
2014-11-28 11:36:14, Info
71), objectname [6]"(null)"
2014-11-28 11:36:14, Info
result 0x00000000, handle @0x4e5c
2014-11-28 11:36:14, Info
Beginning NT transaction commit...
2014-11-28 11:36:14, Info
trace:
CSIPERF:TXCOMMIT;375581

```

generated



7.

Streaming

oThe data flows into the system when an event happens instead of being loaded into a data store in a batch.

